Bellabeat Smart Device Usage Analysis

- **The Business Task:** Analyze competitor FitBit data to identify trends in how consumers use smart fitness devices, and use those insights to recommend data-driven marketing strategies for the Bellabeat app.
- **Key Stakeholders:** Urška Sršen (Co-founder and Chief Creative Officer) and Sando Mur (Co-founder and Executive Team).

Phase 1: Ask

Data Source: FitBit Fitness Tracker Data (Kaggle) — 30 FitBit users, April-May 2016.
Provided by Motivate International Inc. under public license.

Phase 2: Prepare

Data Source

- **Dataset**: FitBit Fitness Tracker Data
- **Source**: Kaggle (Mobius)
- **License**: Public domain, Motivate International Inc.
- **Files used:** dailyActivity_merged.csv,
  sleepDay_merged.csv,
  hourlySteps_merged.csv

Data Credibility Assessment (ROCCC)

- **Reliable**: Low — only 33 users, self-reported data
- **Original**: No — third-party collected via Amazon Mechanical Turk
- **Comprehensive**: Moderate — covers activity, sleep, hourly steps
- **Current**: No — collected April–May 2016, nearly a decade old
- **Cited**: Yes — publicly available under open license

Data Limitations

- Sample size of 33 is below statistical significance threshold
- Data is from 2016 and may not reflect current device behavior
- No demographic data (age, gender, location) available
- Sleep data only available for 24 of 33 users (72.7% coverage)
- Self-selection bias possible — motivated users may over-represent

Despite these limitations, the dataset provides sufficient directional insights to support strategic recommendations.

Phase 3: PROCESS & EXPLORE

Tool: MySQL 8.0 via MySQL Workbench

Approach: SQL used to validate data integrity, establish benchmarks, and answer 8 structured business questions on the dailyActivity dataset.

1.1. Total Number of Unique Users

```sql
-- ================================================
-- SECTION 1: USER & DATA OVERVIEW
-- ================================================
-- 1.1. How many unique users are in the daily activity dataset?
-- Business Purpose: To verify our sample size before deep analysis.
-- ================================================

SELECT COUNT(DISTINCT Id) AS total_users
FROM dailyactivity_merged;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_users
33

Insight: The dataset contains activity data for 33 unique users. While this is a small sample size, it is sufficient for identifying directional trends in how consumers use smart devices.

1.2. Timeframe of the Dataset

```sql
-- ================================================
-- 1.2. What is the start date, the end date, and the total number of days tracked in this dataset?
-- Business Purpose: To understand the exact time frame of the competitor data we are analyzing.
-- ================================================

SELECT
    MIN(STR_TO_DATE(ActivityDate, '%c/%e/%Y')) AS start_date,
    MAX(STR_TO_DATE(ActivityDate, '%c/%e/%Y')) AS end_date,
    COUNT(DISTINCT ActivityDate) AS total_days
FROM dailyactivity_merged;

-- Result:
-- start_date  | end_date   | days_covered
-- 2016-04-12  | 2016-05-12 | 31
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

start_date	end_date	total_days
2016-04-12	2016-05-12	31

Insight: The data covers a 31-day period (roughly one month). This is a solid baseline to identify daily habits, though we should note that a single-month snapshot might not capture long-term seasonal shifts in user behavior.

"Data Limitation: This dataset contains only 33 users over a single month in 2016. Results should be treated as directional trends rather than statistically conclusive findings. A larger, more recent dataset would be needed for definitive business decisions."

1.3. User Tracking Consistency

```sql
-- ==============================================
-- 1.3. How consistently do users track their data?
-- Business Purpose: To see if users wear their devices every day or if they abandon them.
--                   High usage means the app is sticky; low usage means we need better retention features.
-- ==============================================

select id, count(ActivityDate) as total_days
    from dailyactivity_merged
group by id
order by total_days desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	total_days
1503960366	31
1624580081	31
7086361926	31
1844505072	31
1927972279	31
2022484408	31

Result 21 | Read Only

Insight: The top users are highly consistent, tracking their activity for the full 31 days of the study. This indicates strong engagement from power users. Upon reviewing the full results, the minimum tracking time for any user was 4 days, showing that a small segment did abandon the device early.

1.4. Average Daily Steps and Calories

```sql
-- ==============================================
-- 1.4. What are the average daily steps and calories for each user?
-- Business Purpose: To segment users into different lifestyle categories (e.g., highly active vs. sedentary) based on their actual habits.
-- ==============================================

select Id, Round(avg(TotalSteps), 2) as Average_steps,
    round(avg(Calories), 2) as Average_calories
from dailyactivity_merged
group by Id
Order by Average_steps desc, Average_calories desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Id	Average_steps	Average_calories
8877689391	16040.03	3420.26
8053475328	14763.29	2945.81
1503960366	12116.74	1816.42
2022484408	11370.65	2509.97
7007744171	11323.42	2544.00
3977333714	10984.57	1513.67

Insight: There is a significant variance in user habits. Some users are highly active (averaging 12,000+ steps), while others are relatively sedentary (averaging ~2,500 steps). This proves that a "one-size-fits-all" marketing approach won't work; we need to segment our users based on their actual activity levels.

1.5. Average Daily Time by Activity Level

```
-- ================================================
-- 1.5. What is the average daily time spent in different activity levels?
-- Business Purpose:
-- ================================================

select Id, Round(avg(VeryActiveMinutes), 2) as Avg_Veryactiveminutes,
    Round(avg(FairlyActiveMinutes), 2) as Avg_fairlyactiveminutes,
    round(avg(LightlyActiveMinutes), 2) as Avg_lightlyactiveminutes,
    round(avg(SedentaryMinutes), 2) as Avg_sedentaryminutes
from dailyactivity_merged
Group by Id
ORDER BY Avg_Veryactiveminutes DESC, Avg_sedentaryminutes DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Id	Avg_Veryactiveminutes	Avg_fairlyactiveminutes	Avg_lightlyactiveminutes	Avg_sedentaryminutes
5577150313	87.33	29.83	147.93	754.43
8053475328	85.16	9.58	150.97	1148.00
8877689391	66.06	9.94	234.71	1112.87
8378563200	58.68	10.26	156.1	716.13
7086361926	42.58	25.35	143.84	850.45
1503960366	38.71	19.16	219.94	848.16

Result 30 | Read Only

Insight: A significant portion of the user day is spent in the 'Sedentary' state (averaging over 10 hours for many users). However, the top segment of users logs over 60 minutes of 'Very Active' time daily. Bellabeat can leverage this by creating "Sedentary Alerts" to encourage movement during long inactive periods.

1.6. Overall Dataset Benchmarks

```
-- ================================================
-- 1.6. What is the overall average daily steps, calories, and sedentary minutes across ALL users combined?
-- Business Purpose: To establish dataset-wide benchmarks we can compare against WHO health recommendations and use as headline numbers in ou
-- ================================================

select Round(avg(TotalSteps), 0) as overall_avg_steps,
    Round(avg(Calories), 0) as overall_avg_calories,
    Round(avg(SedentaryMinutes), 0) as overall_avg_sedantaryminutes
from dailyactivity_merged

-- Result:
-- overall_avg_steps | overall_avg_calories | overall_avg_sedentarymins
-- 7638              | 2304                 | 991
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

overall_avg_steps	overall_avg_calories	overall_avg_sedantaryminutes
7638	2304	991

Insight: Across the entire dataset, the average user logs 7,638 steps per day, which falls short of the WHO's recommended 10,000 steps. Additionally, the average sedentary time is 991 minutes (over 16 hours) per day. These benchmarks highlight a clear opportunity for Bellabeat to introduce features that specifically target reducing sedentary time and bridging the gap to the 10,000-step goal.

1.7. Users Meeting 10,000 Daily Step Goal

```sql
-- ==============================================
-- 1.7. What percentage of users are meeting the WHO recommended 10,000 steps per day on average?
-- Business Purpose: To quantify the "activity gap" — how many users actually hit the health benchmark vs how many fall short. This becomes t
-- ==============================================

With Useraverages as (
    select Id, Round(avg(TotalSteps), 0) as Avg_steps
    from dailyactivity_merged
    Group by Id),
Userstotal as (
    select count(Id) as total_users,
    sum(case when avg_steps >= 10000 then 1 else 0 end) as users_meeting_goals
    from Useraverages)
select total_users, users_meeting_goals,
    Round((users_meeting_goals / total_users * 100), 1) as pct_users_meeting_goals
from Userstotal;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_users	users_meeting_goals	pct_users_meeting_goals
33	7	21.2

Insight: Only 21.2% (7 out of 33) of users in this dataset are averaging the recommended 10,000 steps per day. This reveals a massive market opportunity for Bellabeat: the vast majority of users (nearly 80%) fall short of basic health benchmarks and need active encouragement, progress tracking, and gamification to reach their goals.

1.8. Average Steps by Day of the Week

MySQL Workbench — Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator — SCHEMAS — bellabeat_exploration

```sql
-- ==============================================
-- 1.8. Which day of the week has the highest and lowest average steps?
-- Business Purpose: To identify peak and low activity days. This tells us when users are most vs least engaged with their devices — critical
-- ==============================================

select dayname(str_to_date(ActivityDate, '%c/%e/%Y')) as day_of_week,
    Round(avg(TotalSteps), 0) as avg_steps
from dailyactivity_merged
group by day_of_week
order by avg_steps desc;

-- Result:
-- day_of_week | avg_steps
-- Saturday    | 8153
-- Tuesday     | 8125
-- Monday      | 7781
-- Wednesday   | 7559
-- Friday      | 7448
-- Thursday    | 7406
-- Sunday      | 6933
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

day_of_week	avg_steps
Saturday	8153
Tuesday	8125
Monday	7781
Wednesday	7559
Friday	7448
Thursday	7406

Result 13 — Read Only — Context Help — Snippets

Insight: Users are most active on Saturdays (8,153 steps) and Tuesdays (8,125 steps), but activity drops significantly on Sundays (6,933 steps). Bellabeat can use this trend to schedule motivational push notifications on Sunday mornings to prevent the end-of-week slump.

Phase 4: ANALYZE

Tool: Python 3 via Google Colab
Libraries: pandas, matplotlib

Datasets: dailyActivity, sleepDay, hourlySteps

Approach: Deep-dive analysis across 14 questions covering activity patterns, sleep behavior, hourly trends, and cross-dataset correlation.

- **Data Preparation & Cleaning (Questions 1, 2, & 3)**

```
Duplicated rows: 0

Nulls values per column:
Id                          0
ActivityDate                0
TotalSteps                  0
TotalDistance               0
TrackerDistance             0
LoggedActivitiesDistance    0
VeryActiveDistance          0
ModeratelyActiveDistance    0
LightActiveDistance         0
SedentaryActiveDistance     0
VeryActiveMinutes           0
FairlyActiveMinutes         0
LightlyActiveMinutes        0
SedentaryMinutes            0
Calories                    0
dtype: int64
```

Insight: The daily activity dataset was successfully loaded and validated. The data is perfectly clean with zero duplicates and zero missing values across all 940 rows, ensuring the accuracy of all following calculations.

Q4: Overall Daily Averages

```
avg_steps = daily_activity['TotalSteps'].mean().round(0)
avg_calories = daily_activity['Calories'].mean().round(0)
avg_sedentary = daily_activity['SedentaryMinutes'].mean().round(0)

print(f"Average Steps: {avg_steps}")
print(f"Average Calories: {avg_calories}")
print(f"Average Sedentary Minutes: {avg_sedentary}")
```

```
Average Steps: 7638.0
Average Calories: 2304.0
Average Sedentary Minutes: 991.0
```

Insight: Python confirms our initial SQL findings: the average user takes 7,638 steps per day, which falls short of the WHO's 10,000-step recommendation. Furthermore, users spend an average of 991 minutes (16.5 hours) per day in a completely sedentary state. This reveals the core inactivity problem Bellabeat must address.

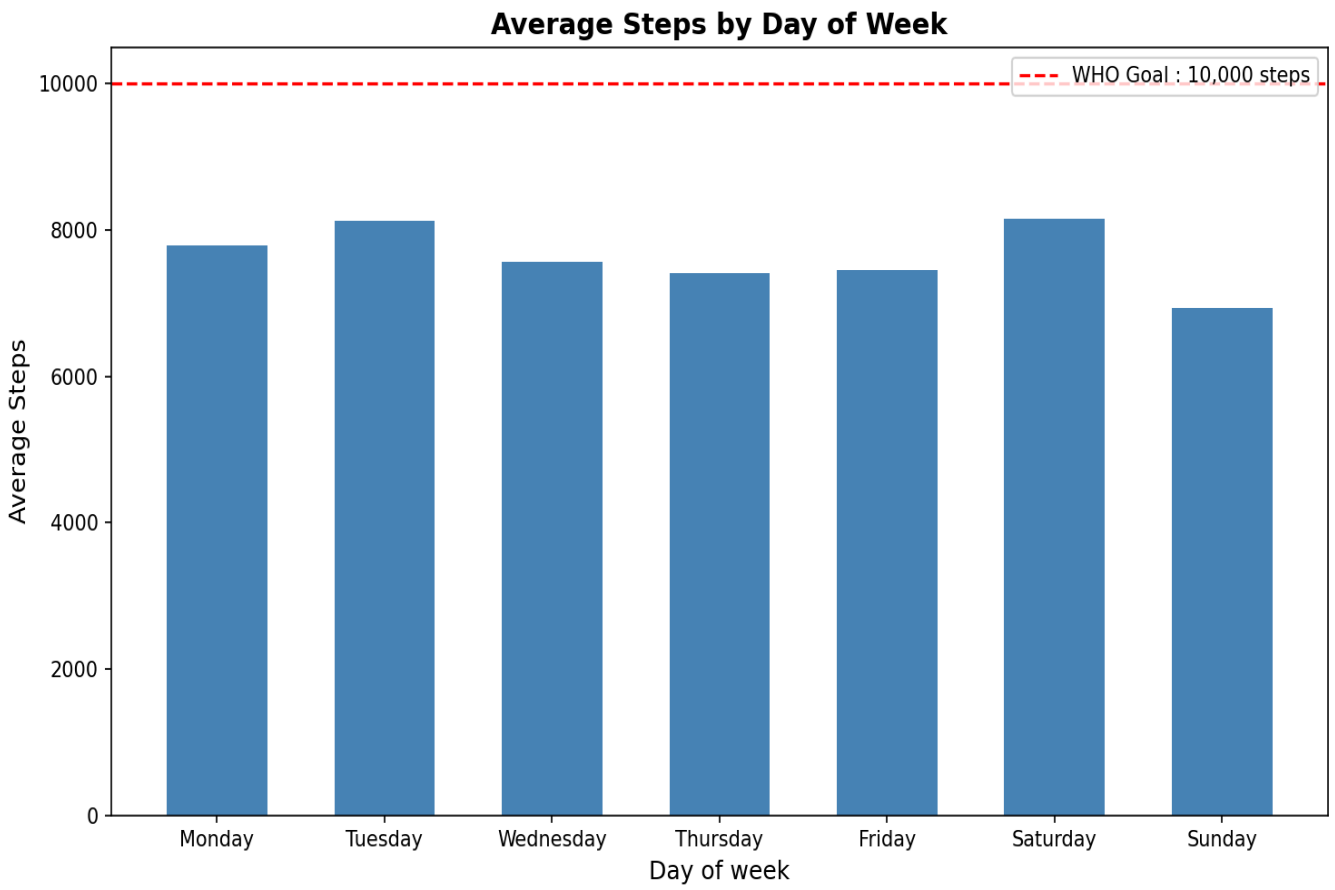
Q5: User Segmentation

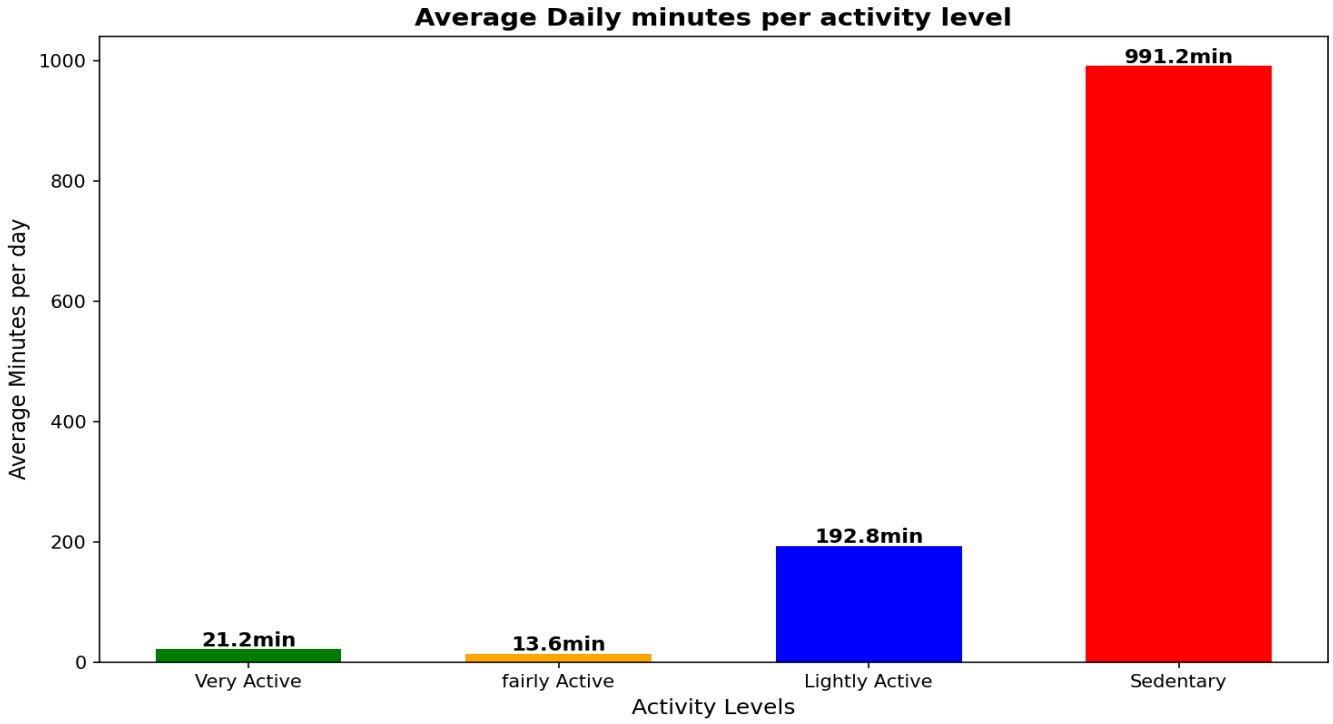
```python
#step 1: get average steps for user
user_avg_steps = daily_activity.groupby('Id')['TotalSteps'].mean().reset_index()
user_avg_steps.columns = ['Id', 'avg_steps']
user_avg_steps['avg_steps'] = user_avg_steps['avg_steps'].round(0)

#step 2: Classify each user
def classify_user(steps):
  if steps >= 10000:
    return 'Active'
  elif steps >= 7500:
    return 'Moderate'
  elif steps >= 5000:
    return 'Light'
  else:
    return 'Sedentary'

user_avg_steps['user_segmentation'] = user_avg_steps['avg_steps'].apply(classify_user)

#step 3 : COunt per segment
segment_counts = user_avg_steps['user_segmentation'].value_counts()
print(segment_counts)
```

```
user_segmentation
Light        9
Moderate     9
Sedentary    8
Active       7
Name: count, dtype: int64
```

Insight: The user base is evenly split across activity levels: 9 Light, 9 Moderate, 8 Sedentary, and only 7 Active users. This indicates that Bellabeat's marketing should not exclusively target athletes, but rather focus on helping average, everyday users build healthier movement habits.

Q6 & Q7: Daily Activity Patterns





Insight: Users are most active on Saturdays and least active on Sundays. However, as the Activity Level chart highlights, regardless of the day, the overwhelming majority of a user's time is spent completely sedentary. Bellabeat's app needs to focus on breaking up these long blocks of sitting during the workday.

Q8 & Q9: Sleep Data Averages

Q8: Load Sleep Data and Initial Inspection

Loading the `sleepDay_merged.csv` dataset to check its structure and the number of participating users.

Concept: pd.read_csv(), .nunique(), .shape, .head()

```
[10]
sleep_data = pd.read_csv('/content/sleepDay_merged.csv')
print('Sleep_data Shape:', sleep_data.shape)
print('Unique users with sleep data:', sleep_data.nunique())
print('\nFirst five rows of data:')
print(sleep_data.head())
```

```
Sleep_data Shape: (413, 5)
Unique users with sleep data: Id                    24
SleepDay              31
TotalSleepRecords      3
TotalMinutesAsleep   256
TotalTimeInBed       242
dtype: int64

First five rows of data:
           Id              SleepDay  TotalSleepRecords  TotalMinutesAsleep  \
0  1503960366  4/12/2016 12:00:00 AM                  1                 327
1  1503960366  4/13/2016 12:00:00 AM                  2                 384
2  1503960366  4/15/2016 12:00:00 AM                  1                 412
3  1503960366  4/16/2016 12:00:00 AM                  2                 340
4  1503960366  4/17/2016 12:00:00 AM                  1                 700

   TotalTimeInBed
0             346
1             407
2             442
3             367
4             712
```

Q9: Data Transformation - Convert Sleep Minutes to Hours

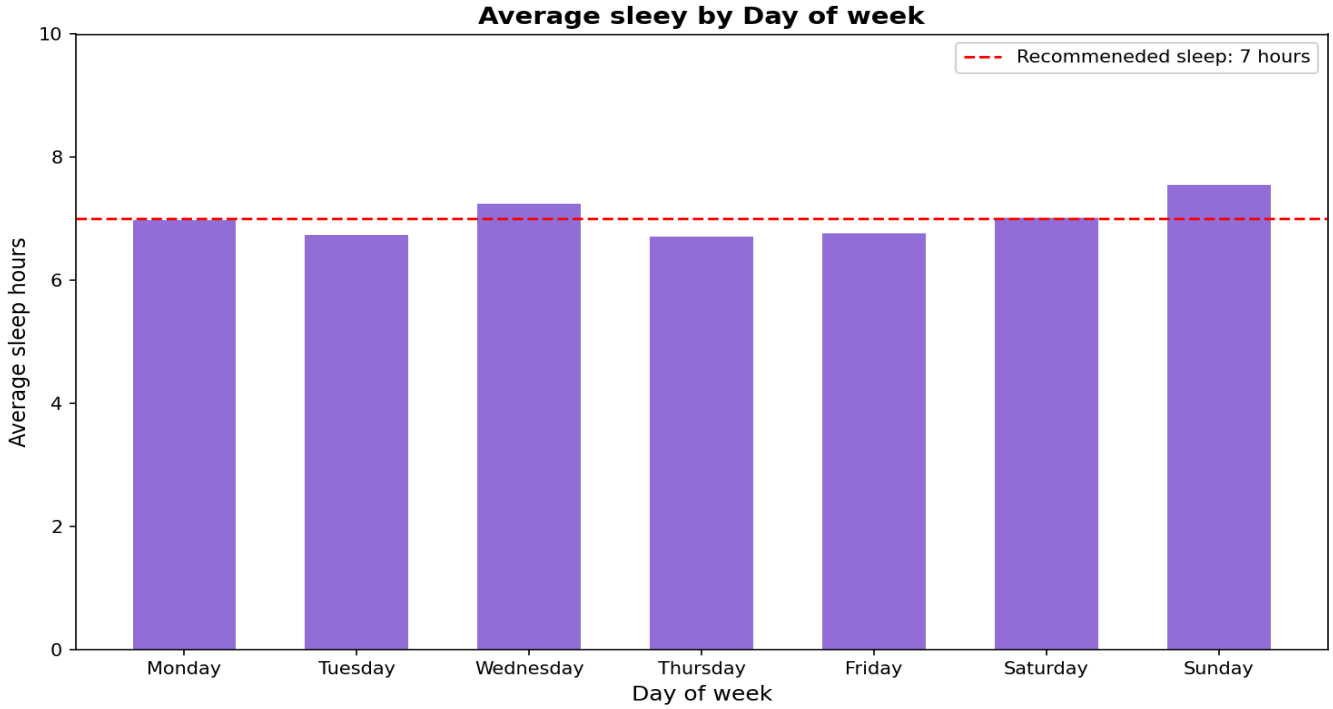
Converting the total minutes asleep into hours to calculate the overall average sleep duration.

```
[11]
sleep_data['sleep_hours'] = sleep_data['TotalMinutesAsleep'] / 60
sleep_data['sleep_hours'] = sleep_data['sleep_hours'].round(2)

avg_sleep = sleep_data['sleep_hours'].mean().round(2)
print(f"Average sleep hours per night: {avg_sleep}")
```

```
Average sleep hours per night: 6.99
```

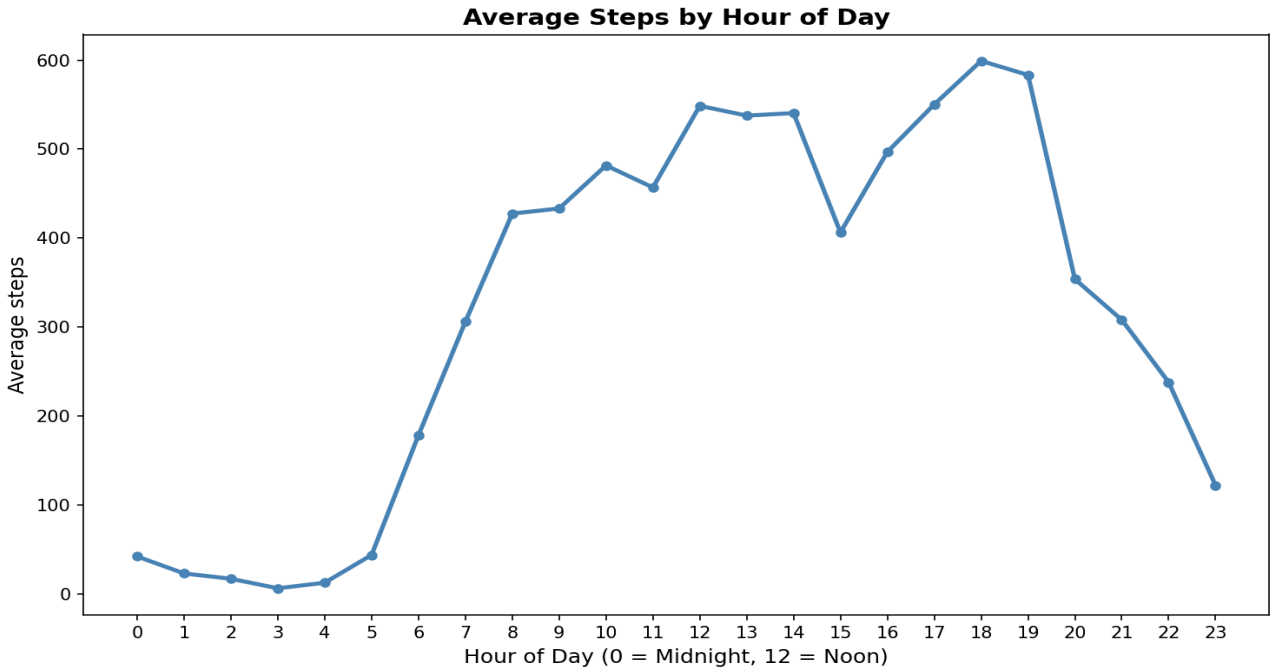
Insight: Sleep data was recorded by 24 of our 33 users. The average sleep duration across all users is 6.99 hours, hovering just below the baseline healthy recommendation of 7 hours.

Q10: Sleep by Day of Week



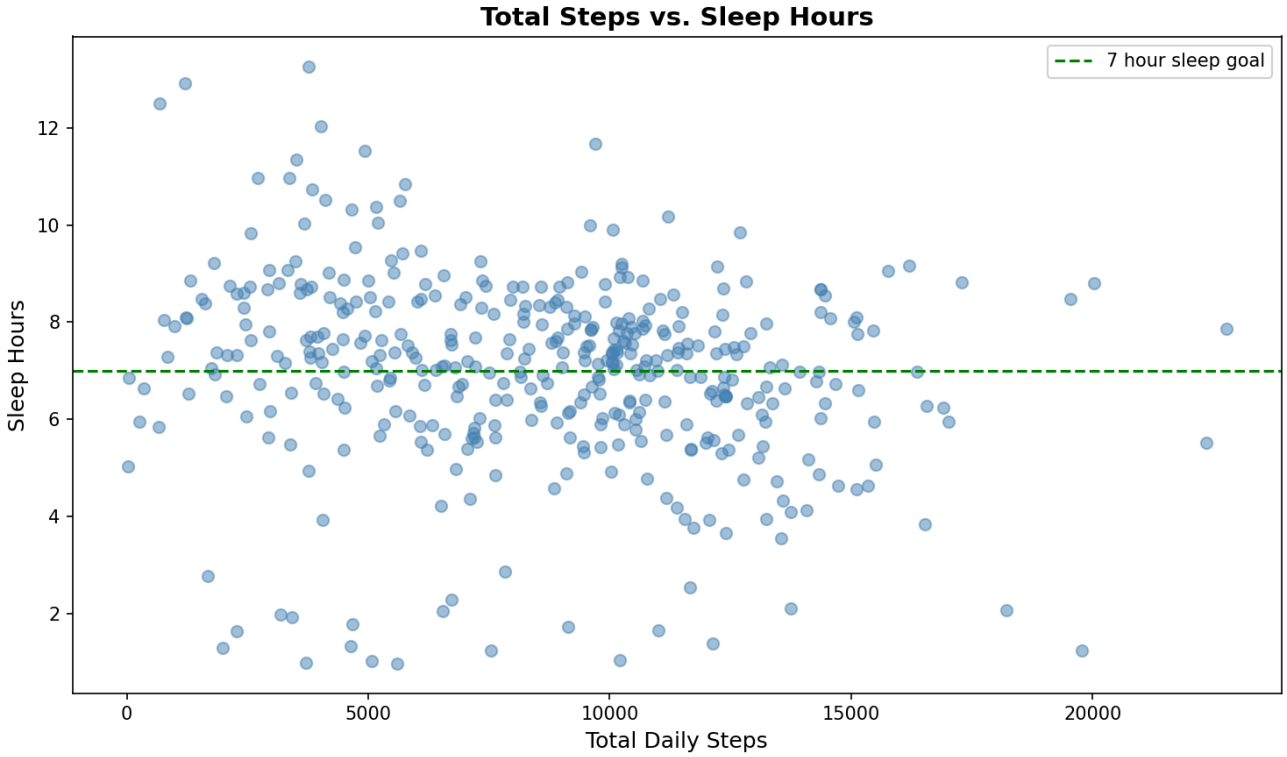
Insight: Users sleep the most on Sundays (the only day averaging above 7 hours), likely using it as a rest day to recover from their highly active Saturdays. Mid-week sleep consistency remains a significant area for improvement.

Q11 & Q12: Hourly Step Trends



Insight: Activity sharply peaks at two specific times: around 12:00 PM (lunch breaks) and between 5:00 PM and 7:00 PM (post-work commutes or workouts). Bellabeat should program their smart device push notifications to trigger exactly at 11:45 AM and 5:45 PM to catch users right before they typically start moving.

Q13 & Q14: Steps vs. Sleep Correlation



Insight: After merging the daily activity and sleep datasets, a scatter plot analysis reveals no strong correlation between the number of steps a user takes and the amount of sleep they get that night. Therefore, Bellabeat should treat daily activity and sleep as two distinct wellness pillars in their app, rather than marketing that walking more will guarantee better sleep.

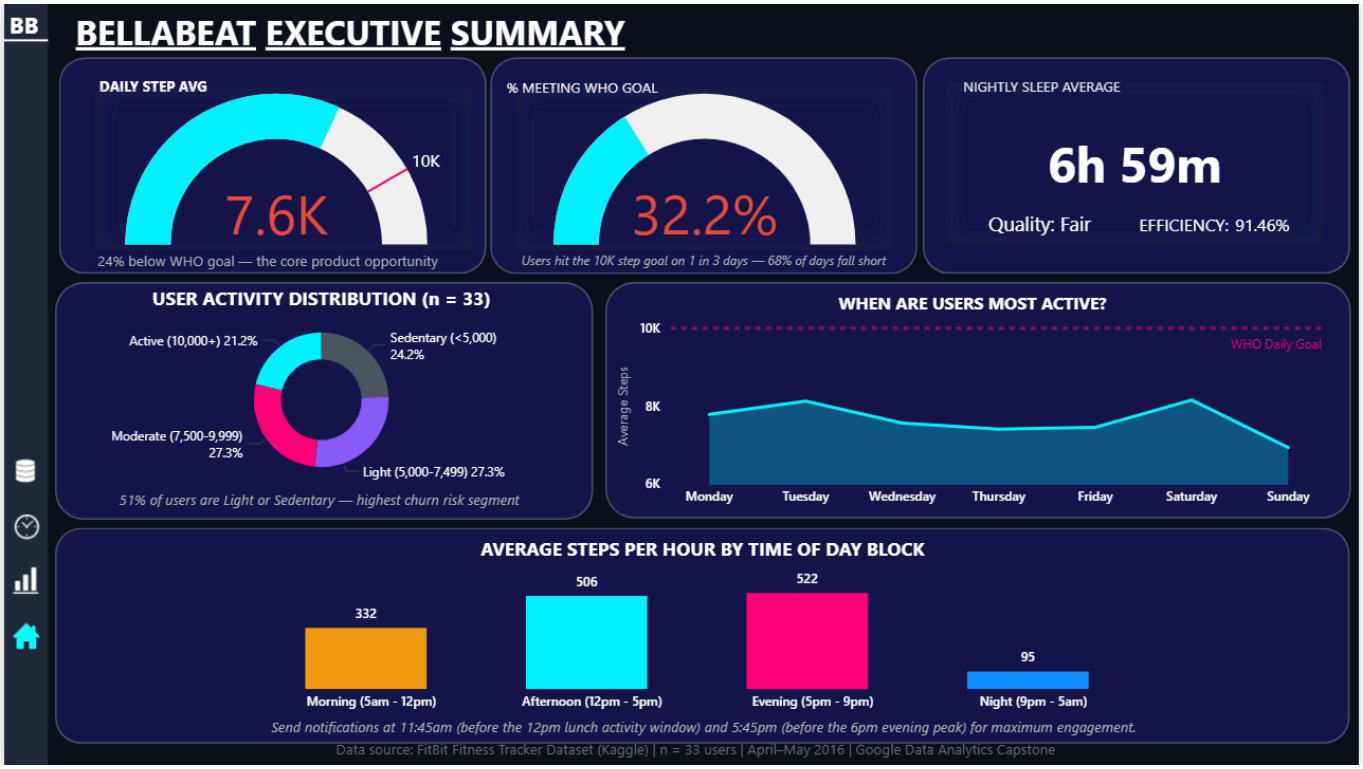
PHASE 5: SHARE — POWER BI DASHBOARD

Tool: Microsoft Power BI Desktop

Live Interactive Dashboard: https://tinyurl.com/bellabeat-ritwik

To translate the raw SQL and Python findings into actionable business strategy, I built a 4-page interactive dashboard tailored for Bellabeat's executive team.

Page 1 — Executive Summary Designed for CEO/CMO level. Three KPI gauges show the core wellness gap. Donut chart segments users by activity level. Activity charts confirm evening as the peak engagement window.

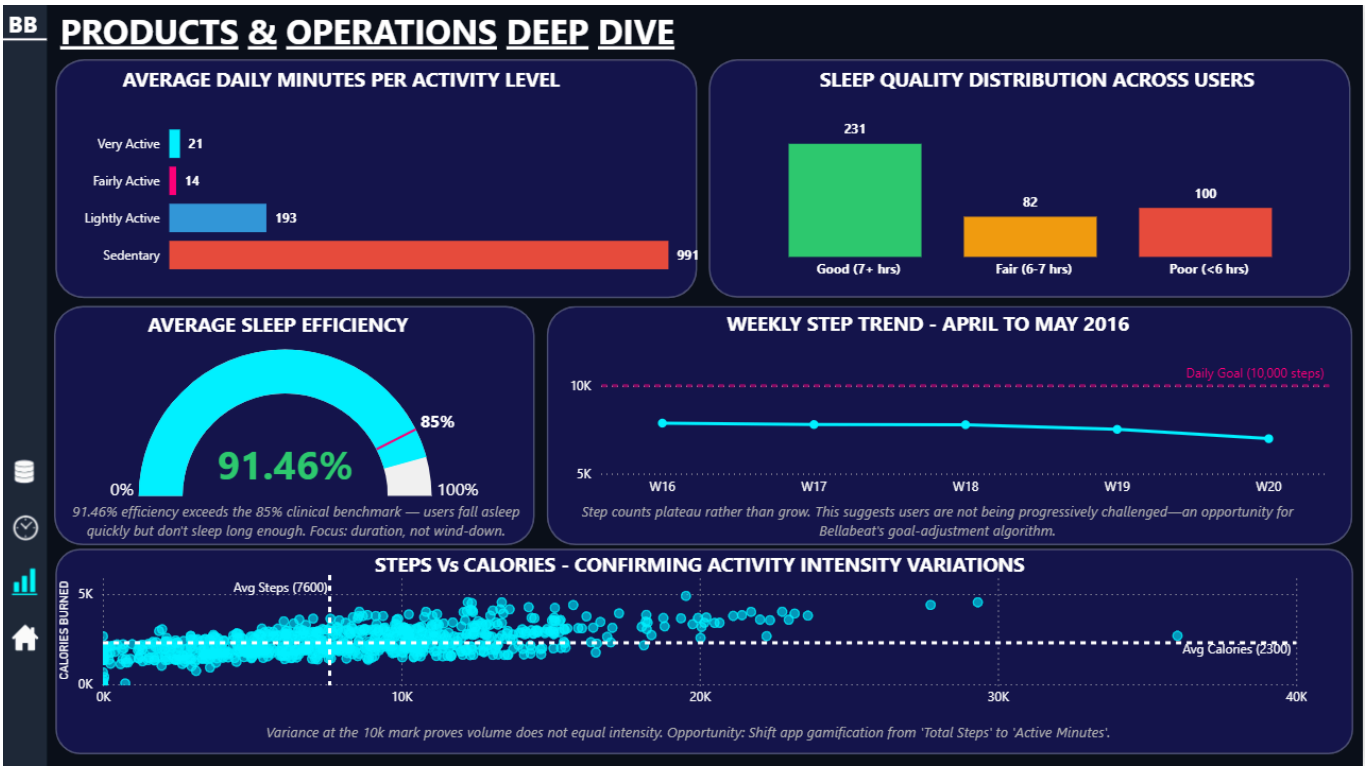


- **The Core Gap:** Users average 7,638 steps/day, falling 24% short of the WHO 10,000 step goal. Only 32.2% of logged days actually hit the target.
- **Sleep Deficit:** Sleep averages 6h 59m. With an efficiency of 91.46%, this proves duration is the primary issue, not sleep quality.
- **The Churn Risk:** 51.5% of users fall into the "Light" or "Sedentary" activity segments, representing the highest risk for platform abandonment.

Page 2 — Manager Dashboard Operational KPIs for product and analytics managers.

Activity breakdown reveals 991 min/day sedentary vs 21 min very active — the 47:1 ratio is the core product problem.

Weekly trend shows plateau, not growth — users need progressive challenges.

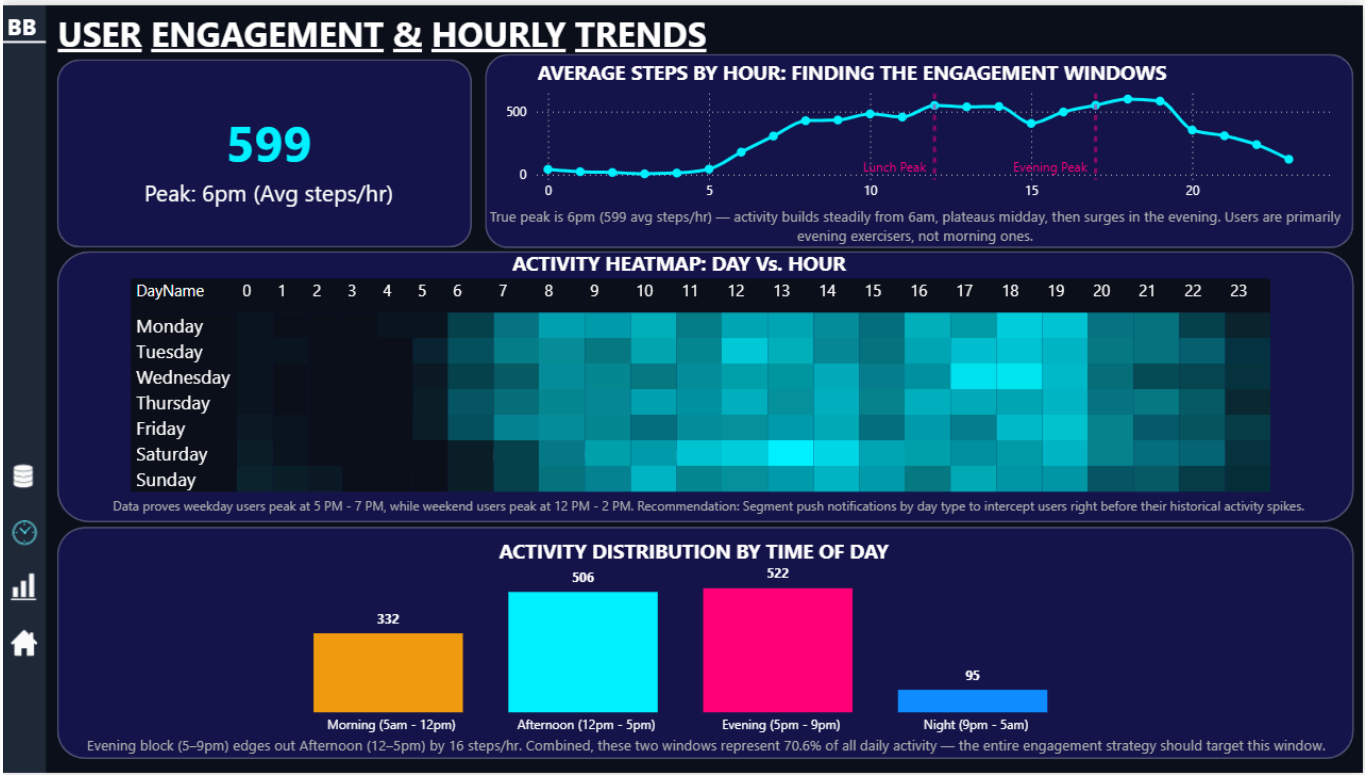


- **Activity Ratio:** Users spend 991 minutes sedentary versus only 21 minutes very active (a 47:1 ratio).
-
- **Gamification Opportunity:** Step counts plateau over time. Bellabeat should shift gamification from "Total Steps" to "Active Minutes" to encourage intensity, not just volume.

Page 3 — User Engagement & Hourly Trends Most differentiated page.

Activity heatmap (Day × Hour) reveals weekday peak at 5–7pm and weekend peak at 12–2pm — enabling day-type-specific notification scheduling.

Evening + Afternoon windows represent 70.6% of all daily activity.

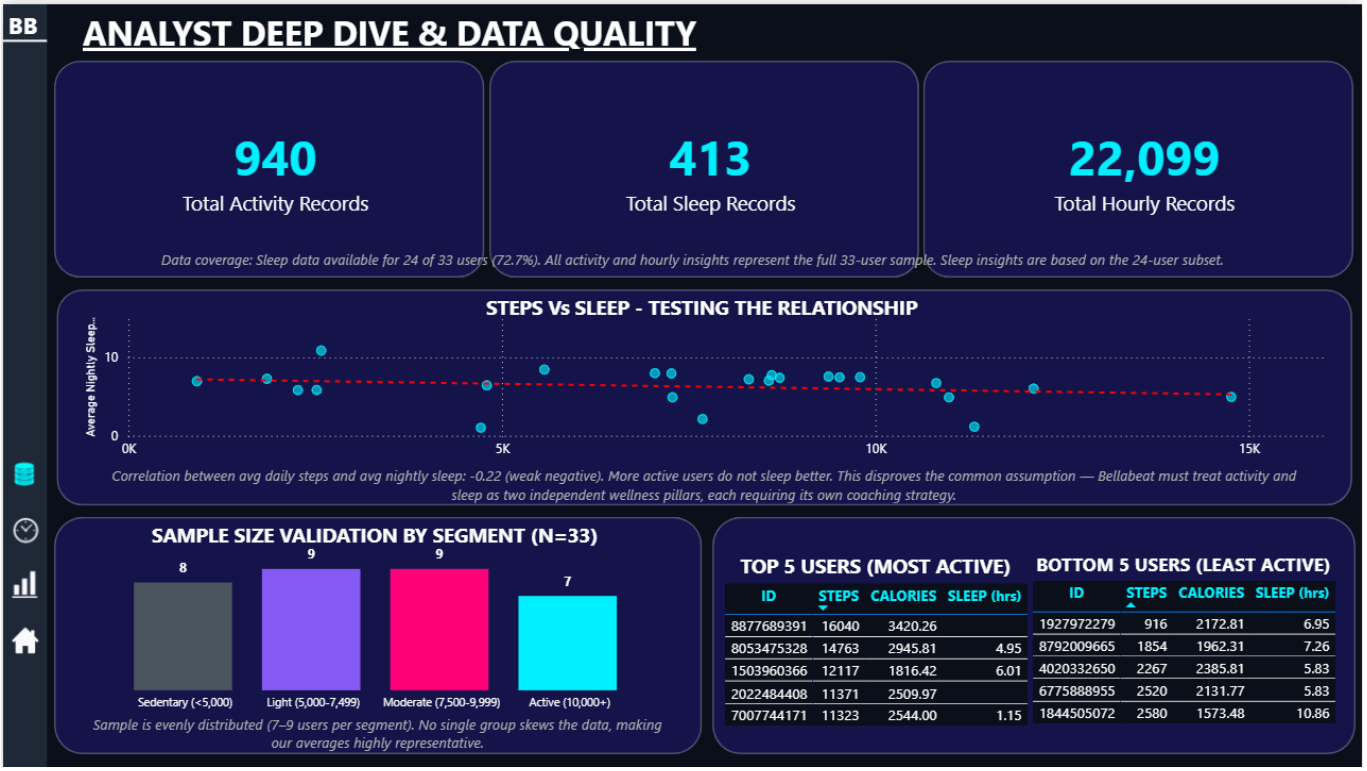


- **Notification Strategy:** A custom activity heatmap reveals peak engagement windows at 6:00 PM on weekdays and 1:00 PM on weekends. Push notifications should be timed 15 minutes prior to these historical spikes.

Page 4 — Analyst Deep Dive & Data Quality

Data quality transparency: 940 activity records, 413 sleep records (24/33 users = 72.7% coverage), 22,099 hourly records.

Steps vs sleep correlation: -0.22 (weak negative) — confirms activity and sleep are independent wellness dimensions requiring separate product strategies.



- **Statistical Conclusion:** Correlation between average daily steps and average nightly sleep is -0.22 (weak negative). This proves more active users *do not* necessarily sleep better, meaning Bellabeat must treat activity and sleep as two independent wellness pillars requiring separate coaching algorithms.

PHASE 6: ACT — STRATEGIC RECOMMENDATIONS

Based on the end-to-end analysis of the Fitbit dataset, Bellabeat should implement the following three product strategies to improve user retention and engagement:

1. **Shift to "Active Minutes" Gamification:** Because step volume does not directly correlate with high-intensity calorie burn (the 47:1 sedentary ratio), the Bellabeat app should introduce "Active Minute" streaks rather than just pushing the standard 10,000 daily step goal.
2. **Deploy Day-Type Smart Notifications:** Activity peaks drastically differ between weekdays (6:00 PM) and weekends (1:00 PM). The marketing and product teams should configure push notifications to trigger exactly 15 minutes before these historical activity spikes to intercept users when they are most likely to exercise.
3. **Decouple Sleep and Activity Coaching:** The data proves a weak negative correlation (-0.22) between steps and sleep duration. Bellabeat must abandon the assumption that "active users sleep better" and build two entirely independent coaching algorithms—one focused on activity volume, and a separate one focused strictly on sleep duration routines.